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import streamlit as st import pickle import numpy as np import pandas as pd import os # -----
----- # Load model and scaler # -----
----- model_path = "heart-disease-model.pkl" if not os.path.exists(model_path): st.error("Model
file not found! Please ensure 'heart-disease-model.pkl' is in the same folder as app.py.")
st.stop() with open(model_path, "rb") as file: model, scaler = pickle.load(file) # -----
----- # App configuration # -----
st.set_page_config(page_title="Heart Disease Prediction", page_icon="❤️", layout="centered")
st.title("🩺 Heart Disease Prediction App") st.write("This app predicts the likelihood of heart
disease based on patient health data.") st.markdown("---") # -----
----- # Define columns (same as used in training) # -----
categorical_cols = [ 'Gender', 'ChestPainType', 'FastingBS', 'RestingECG', 'ExerciseAngina',
'ST_Slope', 'MajorVessels', 'Thalassemia' ] numerical_cols = [ 'Age', 'RestingBp', 'Cholesterol',
'MaxHR', 'ST_Depression' ] # ----- # Collect user inputs # ---
----- col1, col2 = st.columns(2) with col1: Age =
st.number_input("Age", 20, 100, 45) Gender = st.selectbox("Gender", ["Male", "Female"])
ChestPainType = st.selectbox("Chest Pain Type (0:Typical, 1:Atypical, 2:Non-Anginal,
3:Asymptomatic)", [0, 1, 2, 3]) RestingBp = st.number_input("Resting Blood Pressure (mm Hg)",
80, 200, 120) Cholesterol = st.number_input("Cholesterol (mg/dL)", 100, 600, 200) with col2:
FastingBS = st.selectbox("Fasting Blood Sugar > 120 mg/dl", [0, 1]) RestingECG =
st.selectbox("Resting ECG Results", [0, 1, 2]) MaxHR = st.number_input("Maximum Heart Rate
Achieved", 60, 220, 150) ExerciseAngina = st.selectbox("Exercise-Induced Angina", [0, 1])
ST_Depression = st.number_input("ST Depression", 0.0, 6.0, 1.0, step=0.1) ST_Slope =
st.selectbox("ST Slope", [0, 1, 2]) MajorVessels = st.selectbox("Major Vessels (0–3)", [0, 1, 2, 3])
Thalassemia = st.selectbox("Thalassemia (1–3)", [1, 2, 3]) # -----
----- # Preprocess input # ----- Gender = 1 if Gender ==
"Male" else 0 input_dict = { 'Age': Age, 'Gender': Gender, 'ChestPainType': ChestPainType,
'RestingBp': RestingBp, 'Cholesterol': Cholesterol, 'FastingBS': FastingBS, 'RestingECG':
RestingECG, 'MaxHR': MaxHR, 'ExerciseAngina': ExerciseAngina, 'ST_Depression':
ST_Depression, 'ST_Slope': ST_Slope, 'MajorVessels': MajorVessels, 'Thalassemia':
Thalassemia } input_df = pd.DataFrame([input_dict]) input_encoded =
pd.get_dummies(input_df, columns=categorical_cols, drop_first=True) expected_features =
model.feature_names_in_ input_encoded =
input_encoded.reindex(columns=expected_features, fill_value=0) # ensures the DataFrame
columns exactly match the model's training features input_encoded[numerical_cols] =
scaler.fit_transform(input_encoded[numerical_cols]) if st.button("🩺 Predict Heart Disease
Risk"): prediction = model.predict(input_encoded)[0] if prediction == 1: st.error(" High risk of
Heart Disease detected! Please consult a cardiologist.") else: st.success(" No signs of Heart
Disease detected.") st.markdown("---") st.caption("Mini_Project")

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